Faint Bottom Echo Detection for Airborne LiDAR Bathymetry Based on a Constrained Waveform Stacking Model

Xiaolin Yu, Dianpeng Su, Fanlin Yang, Wenze Yang, Zhenhui Wang, Chunxiao Wang, Anxiu Yang, and Chao Qi

***Abstract*—Airborne LiDAR bathymetry (ALB) is particularly suited for acquiring 3-D spatial information in shallow-water zones. However, its signal undergoes rapid attenuation in the water column, resulting in faint bottom echoes that are challenging to detect and directly limit the maximum bathymetric performance. To achieve the accurate extraction of these echoes, a faint bottom echo detection method based on a constrained waveform stacking model is proposed. The model employs distances between water-surface points and their refraction vectors as dual constraints to filter spatially adjacent waveforms. The filtered waveforms are then utilized to construct a pseudo-waveform with a high-signal-to-noise ratio by waveform stacking. On this basis, the accurate faint bottom echo moment is subsequently determined based on a normalized peak significance score. The proposed method is validated using two Mapper-20KU datasets acquired from a coastal zone and an inland reservoir in Qingdao, China. The results demonstrate that the method increases the maximum bathymetric depth by approximately 50% in both regions and significantly improves the point density. Compared with the existing topographic data, the root-mean-square errors (RMSEs) of the derived bathymetric points are 0.22 and 0.09 m, respectively. Therefore, the proposed method can significantly enhance the maximum bathymetric depth while ensuring the bathymetric accuracy.**

***Index Terms*—Airborne LiDAR bathymetry (ALB), constrained waveform stacking, faint bottom echo detection, maximum bathymetric depth.**

I. Introduction

AIRBORNE LiDAR bathymetry (ALB) is an advanced marine surveying technique with advantages such as high efficiency, high point density, and good flexibility [1], [2]. It is commonly used for water–land integrated topographic surveys in the coastal area and can obtain 3-D point cloud data by recording the moment of laser echoes from the water surface and bottom [3], which plays a crucial role in coastal investigation, marine ecological protection, water conservancy construction, and water resources management [4], [5], [6], [7]. Notably, the ALB echo detection directly affects the quality of underwater topographic data, especially the maximum depth and minimum depth [8], [9], [10]. It is the difficult part of the ALB data processing. The early report from National Oceanic and Atmospheric Administration (NOAA) pointed out that the faint bottom echoes (see Fig. 1) are easily obscured by systemic and environmental noise [11]. These phenomena limit the ALB maximum penetration depth and are difficult to be avoided [12], [13], [14]. Therefore, it is necessary to focus on algorithm research for accurately detecting ALB faint bottom echo. Currently, existing methods can be summarized into two categories, including echo detection based on a single waveform and echo detection based on multiple adjacent waveforms.

1) *Bottom Echo Detection Based on a Single Waveform*: Guenther proposed a semiempirical method [11]. The method can improve the recognizability of the faint bottom echo by subtracting the extracted calibration waveform from the bathymetric waveforms, while it is not applicable to all ALB bathymetric waveforms. Schwarz et al. [15] developed an expanded peak detection method based on a significance score. The significance value of each peak is calculated by multiplying the designed peak parameters, and then, the peak with the highest significance is identified as the seafloor echo peak. It is essentially a time-domain filtering method. According to a simplified oceanic LiDAR equation, Richter et al. [16] calibrated the echo intensity based on the volume decay coefficient and applied it to detect bottom echoes. Simultaneously, this method also amplified the intensity of noise signals, which hindered the extraction of the true echo moment. In addition, the continuous wavelet transform is commonly utilized for LiDAR echo extraction because it can better describe the local instantaneous characteristics of signals [17], [18]. However, the extraction results are affected by the predefined wavelet parameters. To solve the problem of nonadaptability in wavelet basis functions, Wu et al. [19] employed the empirical mode decomposition (EMD) method to filter out high-frequency noise from the LiDAR waveform.

Received 18 November 2025; revised 7 March 2026; accepted 28 March 2026. Date of publication 3 April 2026; date of current version 10 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42406186 and Grant 42304051, in part by the Project of Shandong Province Higher Educational Youth Innovation Science and Technology Program under Grant 2023KJ088, in part by the Natural Science Foundation of Shandong under Grant ZR2023QD050 and Grant ZR2024QD062, and in part by the Natural Science Foundation of Qingdao under Grant 24-8-4-zrjj-2-jch and Grant 23-2-1-66-zyyd-jch. *(Corresponding author: Dianpeng Su.)*

Xiaolin Yu and Wenze Yang are with the College of Geodesy and Geomatics, Shandong University of Science and Technology, Qingdao 266590, China.

Dianpeng Su, Fanlin Yang, Anxiu Yang, and Chao Qi are with the College of Geodesy and Geomatics, Shandong University of Science and Technology, Qingdao 266590, China, and also with the Key Laboratory of Marine Surveying and Mapping, Ministry of Natural Resources, Qingdao 266590, China (e-mail: sudianpeng@126.com).

Zhenhui Wang is with The Seventh Topographic Surveying Bridge, Ministry of Natural Resources of China, Haikou 570203, China.

Chunxiao Wang is with Hainan Basic Geographic Information Centre, Ministry of Natural Resources of China, Haikou 570203, China.

Digital Object Identifier 10.1109/TGRS.2026.3680599

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

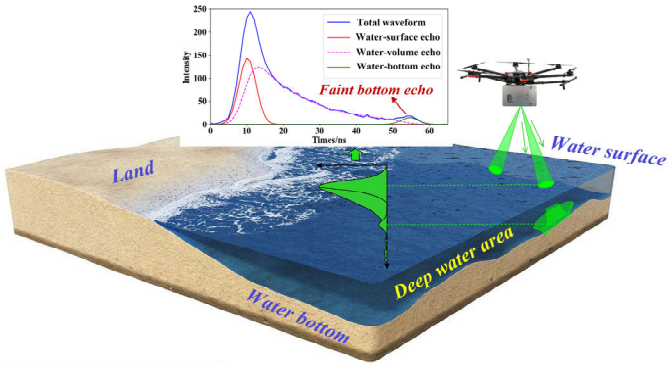


Fig. 1. Illustration of the ALB faint bottom echo.

The EMD method requires specific criteria to distinguish between the intrinsic mode functions of noise and valid echo; thus, Li et al. [20] and Zhou et al. [21] proposed an EMD-based multiresolution analysis method and addressed this issue. Overall, the above methods mainly considered the time–frequency feature differences between noise and target echoes within a single waveform [22], [23], while ignored spatial correlation of the adjacent waveforms.

2) *Bottom Echo Detection Based on Multiple Adjacent Waveforms:* Existing related algorithms mainly utilized the correlation between spatial adjacent waveforms and were designed to improve the effect of faint echo detection by specific constrained conditions [24], [25]. Huang et al. [26] proposed an improved faint bottom echo detection method with PointNet++, which automatically realizes the detection of weak seafloor echo points by learning the features between nearest neighbor waveforms and the points. The applicability of this method is constrained by variations in waveform and topographic characteristics across different water bodies. Yao and Stilla [27] developed a faint pulse detection method combining waveform stacking with principal components analysis. It can effectively recover point data with weak pulse intensity by utilizing the correlation of spatial nearest neighbor waveforms. Similarly, Mader et al. [28] used the waveform stacking method for ALB faint underwater echo detection. The method determined whether the waveforms are adjacent in a grid manner. On this basis, a volumetric nonlinear orthogonal waveform stacking algorithm was proposed [29], [30]. Compared with the RIEGL online waveform processing [31], it can effectively increase the maximum depth and the underwater point density of the ALB. To prevent the insufficient waveform data in the neighborhood unit, the above methods require determining the neighborhood size according to the point density, which limits the flexibility of the methods and reduces their effectiveness for point clouds with nonuniform spatial distributions. Therefore, the faint bottom echo detection method should be further optimized to fully account for correlations between adjacent waveforms.

To enhance the universality of the ALB faint bottom echo detection method, a constrained waveform stacking model is proposed based on water-surface geometric elements, which is verified to effectively increase the ALB maximum depth performance. The rest of this article is structured as follows. Section II offers the study area and data. Section III provides a detailed description of the methodology, which is divided into three parts, including data preprocessing, pseudo-waveform construction, and bottom echo moment determination. Section IV presents the experimental results, along with the corresponding analysis. Finally, the discussion and conclusion are described in Sections V and VI, respectively.

II. Study Area and Data

In this study, two ALB datasets were acquired around Qingdao, with one collected in September 2023 from a coastal zone [Region A, Fig. 2(a)] and the other in July 2025 from an inland reservoir [Region B, Fig. 2(b)]. The former was employed for effectiveness evaluation, and the latter for applicability validation. The water quality in both regions is below case II water environmental standard. In addition, the ALB system used was the Mapper-20KU [see Fig. 2(c)], which was produced by Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences (CAS), and equipped with a dual-wavelength laser of 532 and 1064 nm. The system features a low weight of 6 kg, a 20-kHz repetition rate, and an elliptical scanning pattern with a scanning angle of $\pm 20°$. During the data collection process in the two regions, the flight altitudes were set to 120 and 100 m, with corresponding flight speeds of 23 and 5 m/s.

For quantitative analysis, the reference data are necessary. In Region A, dense reefs precluded vessel access and acoustic depth data acquisition. Fortunately, topographic data collected by the RIEGL VQ-840-G at the same site in October 2022 are available. The data were obtained under low-tide conditions and demonstrated a vertical accuracy better than 5 cm, as verified against 12 measurement points of real-time kinematic (RTK). In Region B, the reference data are multibeam echosounders (MBESs) data acquired in October 2025, and its total vertical uncertainty (TVU) is 19 cm at 10-m water depth, meeting the IHO S-44 Special Order Standard [39].

III. Methods

The spatially adjacent waveforms exhibit similar bottom echo moments. Based on this, the proposed method is designed as shown in Fig. 3 and is outlined as follows. First, the ALB data are preprocessed to extract water-surface points and their corresponding refraction vectors. These two geometric elements are then used to construct a constrained waveform stacking model, producing a pseudo-waveform with an enhanced signal-to-noise ratio. Finally, the faint bottom echo moment is progressively determined using the normalized peak significance score (NPSS), and the corresponding underwater points can be calculated.

A. Data Preprocessing

Data preprocessing is a crucial step before bottom echo detection because it directly affects the quality of the subsequently constructed pseudo-waveform. In this article, it mainly involves determining the water-surface echo moment and obtaining the water-surface points along with their corresponding refraction vectors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

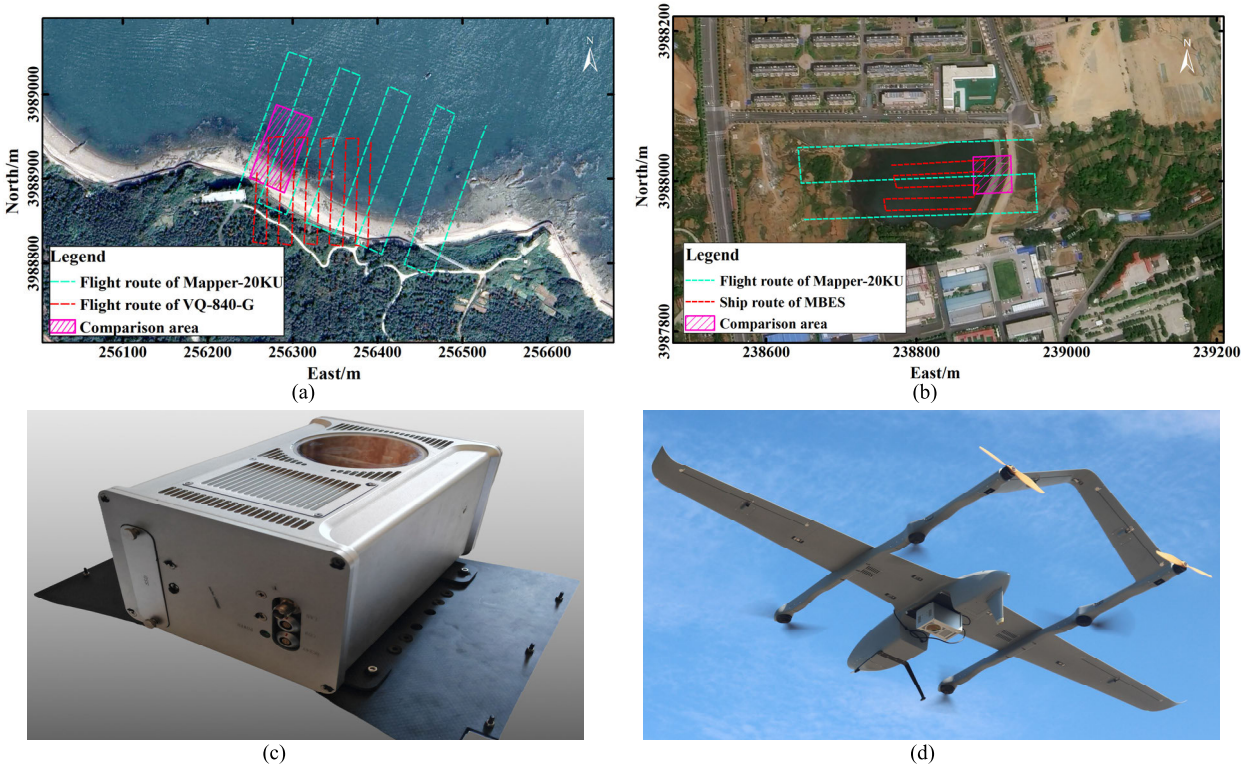


Fig. 2. Experimental regions and equipment. (a) Experimental Region A. (b) Experimental Region B. (c) Mapper-20KU ALB system. (d) Field measurement flight.

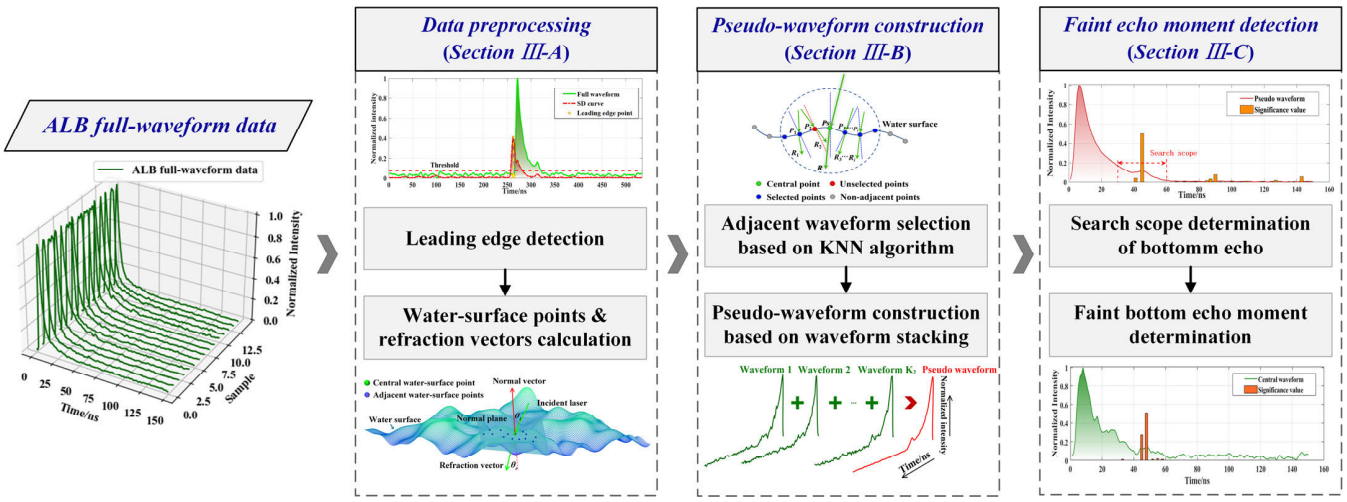


Fig. 3. Workflow of the proposed faint bottom echo detection method.

1) Water-Surface Echo Moment Determination: The water-surface echo moment directly influences waveform stacking, as it serves as the starting position for the process. Therefore, acquiring a stable and precise water-surface echo moment is necessary. However, the intensity of the water-surface echo is affected by factors, such as wind and incident angle [32], [33]. When the intensity is weak, the water-surface echo can mix with the water-volume echo, causing its peak to appear later.

To reduce the impact of the water-volume echo on the water-surface echo moment determination, a leading-edge detection tool based on a sliding window is utilized. It can extract the leading edge according to the standard deviation (SD) difference between the target echo and noise, where the SD value of the leading edge of the target echo changes rapidly, while the noise maintains a stable SD value due to its characteristic as Gaussian white noise [34]. The SD value of the ith echo sample can be calculated within the interval $[i,\ i+n]$ using the following equation:

$$s_i = \mathrm{std}\left(\boldsymbol{w}\left[i : i+n\right]\right) \tag{1}$$

where s_i is the ith SD value; std() represents the calculation of the SD; $\boldsymbol{w}$ is the original waveform; $[i{:}\ i+n]$ is the sliding window, and the window length n is equal to the leading-edge length of the ALB system waveform and is set to 5 in this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

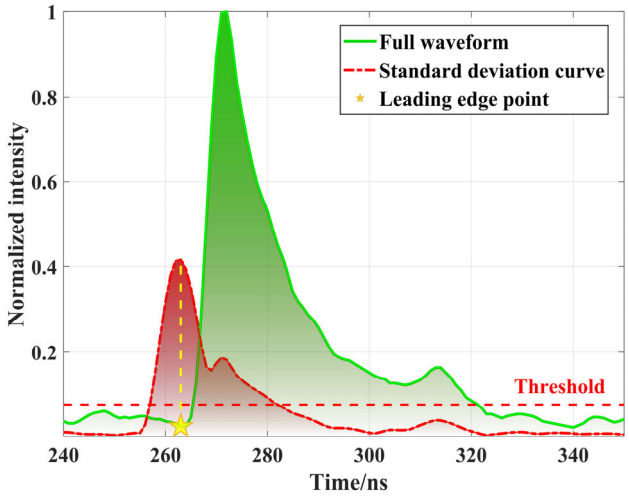


Fig. 4. Determination of water-surface echo moment based on leading-edge detection.

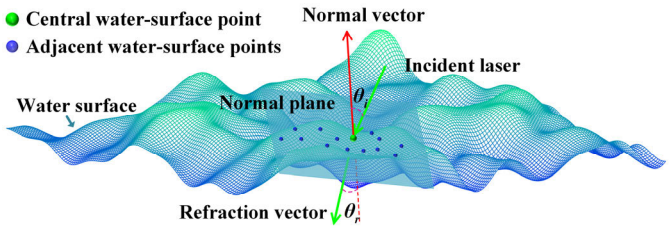


Fig. 5. Laser refraction correction at the air–water interface.

article. Then, the SD curve of each waveform can be generated (see Fig. 4). The SD value of the leading edge shows a peak within the curve. Therefore, the position of the leading edge can be obtained using the following equation:

$$\begin{cases} t_s = \text{find}\left(\boldsymbol{S}\left[\boldsymbol{T_e}\right] > \varepsilon_s\right)[0] \\ \boldsymbol{T_e} = \text{find}\left(\text{diff}\left(\text{sign}\left(\text{diff}\left(\boldsymbol{S}\right)\right)\right) < 0\right) + 1 \end{cases} \tag{2}$$

where t_s is the water-surface echo moment; $\boldsymbol{S} = [s_1, s_2, \ldots, s_m]$ is the SD curve vector; $\boldsymbol{T_e}$ is the position vector of the peaks in curve $\boldsymbol{S}$; ε_s is the noise threshold; find(), diff(), and sign() represent the position search, differential operation, and sign function, respectively. After that, the water-surface echo moment can be determined, and the corresponding water-surface point can be acquired.

2) Water–Air Interface Refraction Correction: The water surface is always in a state of dynamic change, affected by factors such as tides and sea breezes. Therefore, when the laser enters the water volume, the local water–air interface may not remain horizontal (see Fig. 5). This condition will cause a deviation in the laser refraction path and a refraction error at the bottom point [35], [36]. In this article, the adaptive neighborhood-based principal component analysis is employed to derive the water-surface normal vector [37], [38]. It can adaptively select the appropriate neighborhood radius with the minimum entropy, and the first principal component vector at this radius is considered as the water-surface normal vector $\boldsymbol{N_i}$. Then, the refractive vector $\boldsymbol{R_i}$ corresponding to the central water-surface point $\boldsymbol{P_{si}}$ can be computed by the following equation:

$$\begin{cases} \boldsymbol{R_i} = \dfrac{n_a}{n_w}\boldsymbol{I_i} + \left(\dfrac{n_a}{n_w}\cos\theta_i - \sqrt{1-\sin^2\theta_r}\right)\boldsymbol{N_i} \\ \sin^2\theta_r = \left(\dfrac{n_a}{n_w}\right)^2\left(1-\cos^2\theta_i\right) \end{cases} \tag{3}$$

where $\boldsymbol{I_i}$ is the incident vector, n_a and n_w are the optical refractive indices of air and water, respectively, and θ_i and θ_r are the incident and refracted angles, respectively.

B. Pseudo-Waveform Construction Based on the Constrained Waveform Stacking Model

Based on the water-surface points $\boldsymbol{P_s}$ and the refraction vectors $\boldsymbol{R}$ obtained above, a constrained waveform stacking model is constructed to acquire the pseudo-waveform with a higher signal-to-noise ratio. This model consists of two components: the selection of spatially adjacent waveforms and the construction of the pseudo-waveform. Waveform stacking requires spatially adjacent waveform data. Therefore, the criteria for selecting adjacent data are an important factor affecting the quality of the pseudo-waveform. In this section, the K-nearest neighbor (KNN) method is employed to get adjacent waveforms, with the criteria based on the Euclidean distance between water-surface points and refraction vectors. The details are as follows.

1) The Euclidean distance between the central point $\boldsymbol{P_{si}}$ and each other water-surface point is calculated, then the vector set $\Delta\boldsymbol{P_{si}} = [\Delta P_{si1}, \Delta P_{si2}, \ldots, \Delta P_{sim}]$ can be acquired, where ΔP_{sij} is equal to $\|\boldsymbol{P_{si}} - \boldsymbol{P_{sj}}\|^2$, and $\|\ \|^2$ represents the L_2 norm.

2) Then, the water-surface point set $\boldsymbol{P_{si}^n}$, closest to $\boldsymbol{P_{si}}$, and the corresponding refraction vector set $\boldsymbol{R_i^n}$ can be first obtained using KNN, whose K value is set to K_1. It can be written as follows:

$$\begin{cases} \boldsymbol{P_{si}^n} = \boldsymbol{P_s}\left[\mathbf{IDX}_1\right] \\ \boldsymbol{R_i^n} = \boldsymbol{R}\left[\mathbf{IDX}_1\right] \\ \mathbf{IDX}_1 = \text{argsort}\left(\Delta\boldsymbol{P_{si}}\right)[:K_1] \end{cases} \tag{4}$$

where $\mathbf{IDX}_1$ is the index vector of the first-filtered adjacent water-surface points and argsort() is the ascending sorting function.

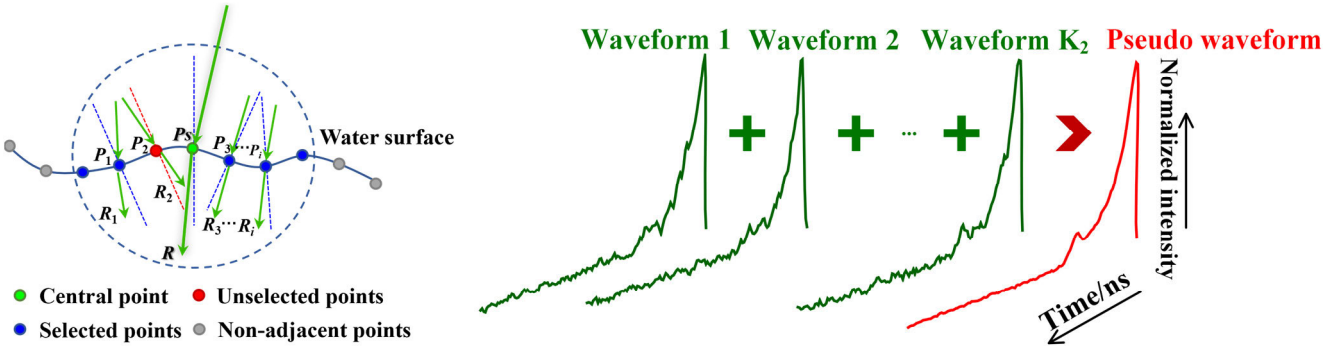
3) The refraction vector set $\boldsymbol{R_{si}^f}$, closest to $\boldsymbol{R_i}$, is refiltered from $\boldsymbol{R_i^n}$. Then the corresponding water-surface point set $\boldsymbol{P_{si}^f}$ is obtained, as shown in the following equation:

$$\begin{cases} \boldsymbol{P_{si}^f} = \boldsymbol{P_{si}^n}\left[\mathbf{IDX}_2\right] \\ \boldsymbol{R_i^f} = \boldsymbol{R_i^n}\left[\mathbf{IDX}_2\right] \\ \mathbf{IDX}_2 = \text{argsort}\left(\|\|\boldsymbol{R_i} - \boldsymbol{R_i^n}\|_2\right)[:K_2] \end{cases} \tag{5}$$

where K_2 is the number of refiltered adjacent water-surface points and satisfies the empirical relation $K_1 = 1.5\times K_2$. The optimal value of K_2 is determined in Section V; $\mathbf{IDX}_2$ is the index vector of the refiltered adjacent water-surface points.

4) Finally, the spatial adjacent waveform set $\boldsymbol{W_i}$ corresponding to the selected water-surface point set $\boldsymbol{P_{si}^f}$ can be identified [see Fig. 6(a)], and a pseudo-waveform can be constructed by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 6. Pseudo-waveform construction using the constrained waveform stacking model. (a) Selected water-surface points corresponding to the adjacent waveforms. (b) Waveform stacking.

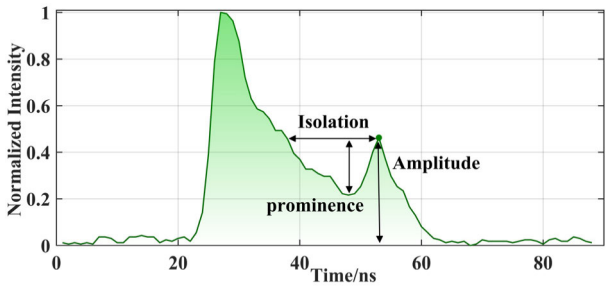


Fig. 7. Three peak parameters for NPSS calculation.

stacking all waveforms of the set $\boldsymbol{W_i}$ [see Fig. 6(b)]. It can be expressed as follows:

$$\omega_{acc} = \sum_{i=1}^{K_2} \omega_i[t_{si} : t_{si} + l] \tag{6}$$

where ω_{acc} is the pseudo-waveform, t_{si} is the water-surface echo moment of the ith adjacent waveform, and l is the length of the truncated waveform.

In this model, the Euclidean distances between water-surface points and refraction vectors serve as the constraint criteria for adjacent waveforms selection. The former can ensure the proximity of the laser position when incident on the water body, and the latter can ensure the similarity of the laser transmission direction. Therefore, this method can mitigate bottom echo broadening in the pseudo-waveform caused by refraction and topographic variations, which is beneficial for subsequent bottom echo moment determination.

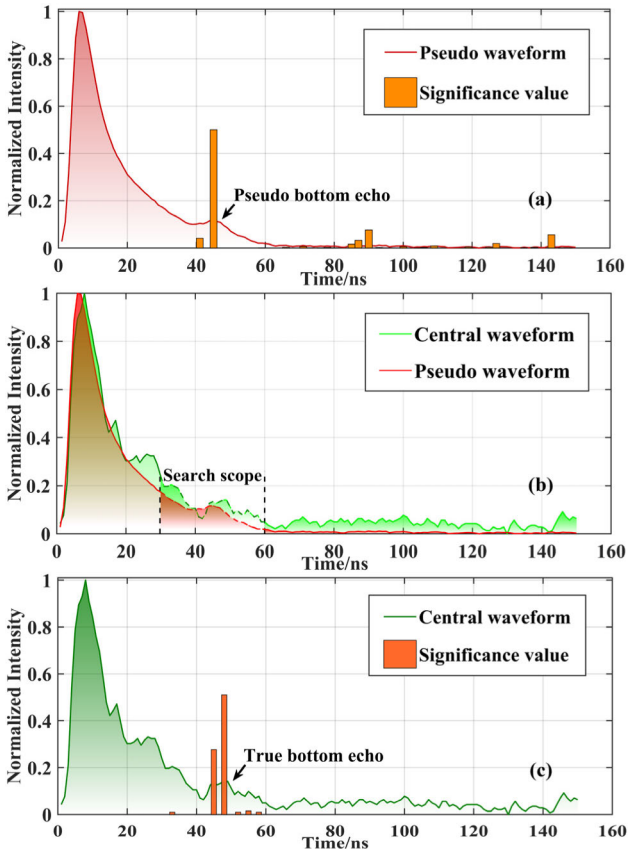


Fig. 8. Bottom echo moment determination. (a) Approximate bottom echo moment determination. (b) Search scope determination. (c) Final bottom echo moment determination.

C. Bottom Echo Moment Determination Based on Normalized Peak Significance Score

The waveform noise is high-frequency Gaussian white noise, characterized by a constant statistical expectation, whereas the target echo appears as a low-frequency signal concentrated within a time range. Therefore, by stacking adjacent waveforms, noise can be reduced, yielding a pseudo-waveform that enhances the bottom echo. Nevertheless, due to variations in the water-surface incidence points, refraction angles, and bottom point positions corresponding to each laser beam, the echo moment varies within a certain range. This leads to broadening of the bottom echo after stacking, such that its moment in the pseudo-waveform cannot represent the true echo moment of the central waveform. To address this issue, based on Schwarz's research, an improved NPSS is proposed to accurately extract the actual bottom echo moment [15].

The improved NPSS value is the result of multiplying the three peak parameters, as expressed in the following equation:

$$\mathrm{NPSS} = \mathrm{P}' \times \mathrm{I}' \times \mathrm{A}' \tag{7}$$

where P′, I′, and A′ are the normalized forms of prominence, isolation, and amplitude, respectively (see Fig. 7). This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

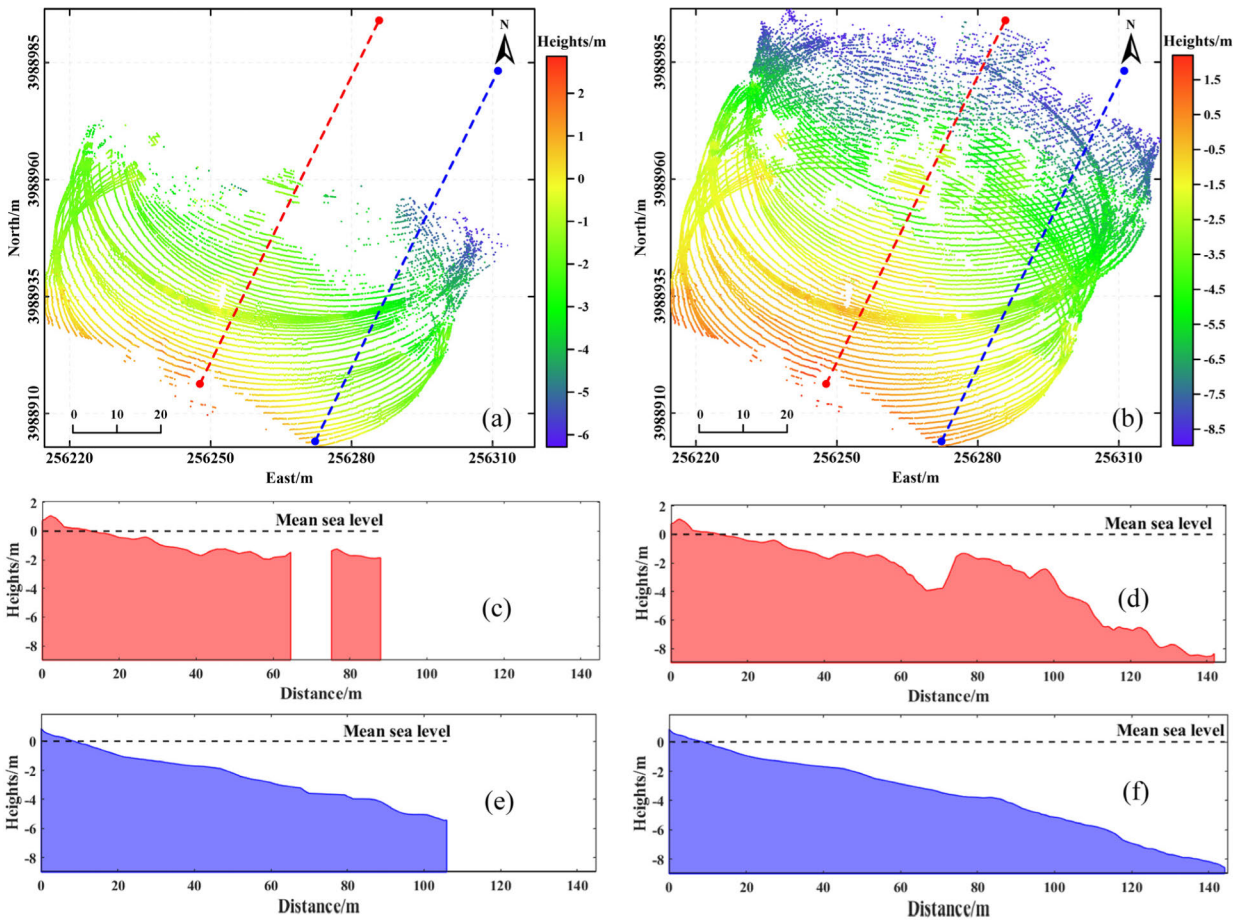


Fig. 9. Analysis of maximum depth in comparison area of Region A. (a) Topographic point cloud obtained by the SPD method. (b) Topographic point cloud obtained by the proposed method. (c) Profile corresponding to the red dashed line in Fig. 9(a). (d) Profile corresponding to the red dashed line in Fig. 9(b). (e) Profile corresponding to the blue dashed line in Fig. 9(a). (f) Profile corresponding to the blue dashed line in Fig. 9(b).

normalization is applied to each parameter across all peaks to eliminate the influence of differing parameter scales. To prevent the NPSS value from becoming 0, the normalized values are scaled to a range of 0.01–1. Based on this, the bottom echo moment can be determined using the following steps.

1) The echo moment with the largest NPSS value is taken as the approximate pseudo-bottom echo moment t_a [see Fig. 8(a)], and the prominence of this peak needs to be greater than the noise threshold ε_b. The threshold can be computed by the following equation:

$$\begin{cases} \varepsilon_b = 3 \times \text{std}\left(\omega_{acc}^n\right) + \text{mean}\left(\omega_{acc}^n\right) \\ \omega_{acc}^n = \omega_{acc}\left[-s:\right] \end{cases} \tag{8}$$

where ω_{acc}^n is the tail noise in the pseudo-waveform; $\omega_{acc}^n[-s:]$ is the last s values of the pseudo-waveform with s is set to 30; and mean() is the mean function.

2) The search scope is then defined, which serves as the tolerance range for determining the moment of bottom echo, as shown in Fig. 8(b). The scope is twice the time range of the falling edge and symmetrical around the approximate moment t_a. In addition, the time range of the falling edge is the time interval from the approximate bottom echo moment to the maximum noise signal moment.

3) Within the search scope of the central waveform, if the prominence of the peak with the maximum NPSS value exceeds half the amplitude range of the tail noise, the echo moment is considered as the bottom echo moment [see Fig. 8(c)]; otherwise, no echo is considered present.

IV. Experimental Results and Analysis

To evaluate the performance of the proposed method, the maximum depth and accuracy of the experimental results from Region A are mainly analyzed. In Region A, the maximum depth is judged by the instantaneous depth relative to the mean water surface, which refers to the local mean sea level (MSL). In Region B, however, the measured mean water-surface elevation is adopted directly due to the calm water conditions. In addition, the depth accuracy is assessed based on three indexes including bias, root mean square error (RMSE), and R-square (R^2). The three indexes can be calculated from the experimental results and reference data in the comparison areas, which are marked as pink in Fig. 2(a) and (b).

A. Analysis of Maximum Depth

As shown in Fig. 9(b), compared with the results obtained using the SPD method [see Fig. 9(a)], the bathymetric points derived from the proposed method extend further into the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

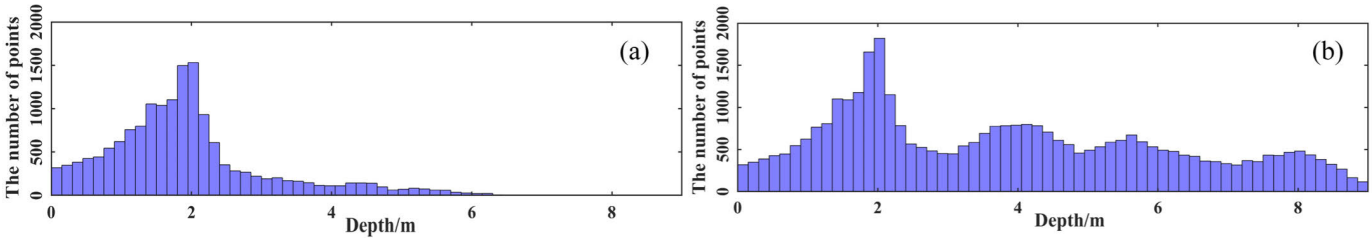


Fig. 10. Number distribution of bathymetric points. (a) Distribution histogram corresponds to Fig. 9(a). (b) Distribution histogram corresponds to Fig. 9(b).

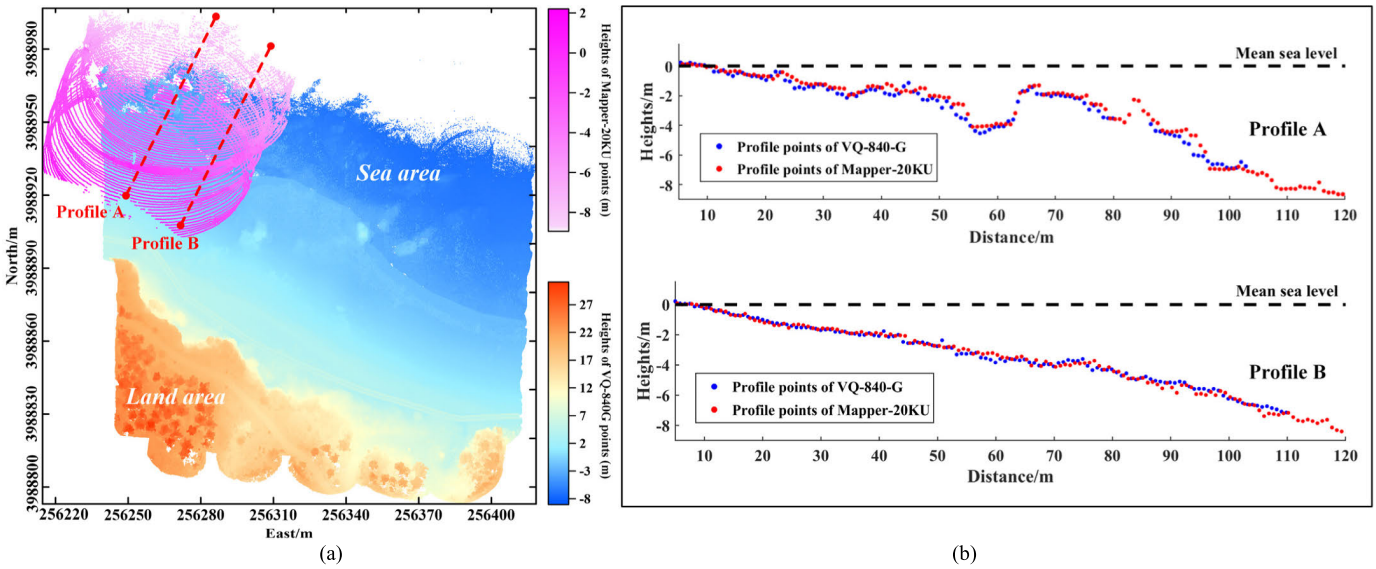


Fig. 11. Comparison of topographic profiles in Region A. (a) Locations of two topographic profiles. (b) Seafloor points corresponding to the two profiles.

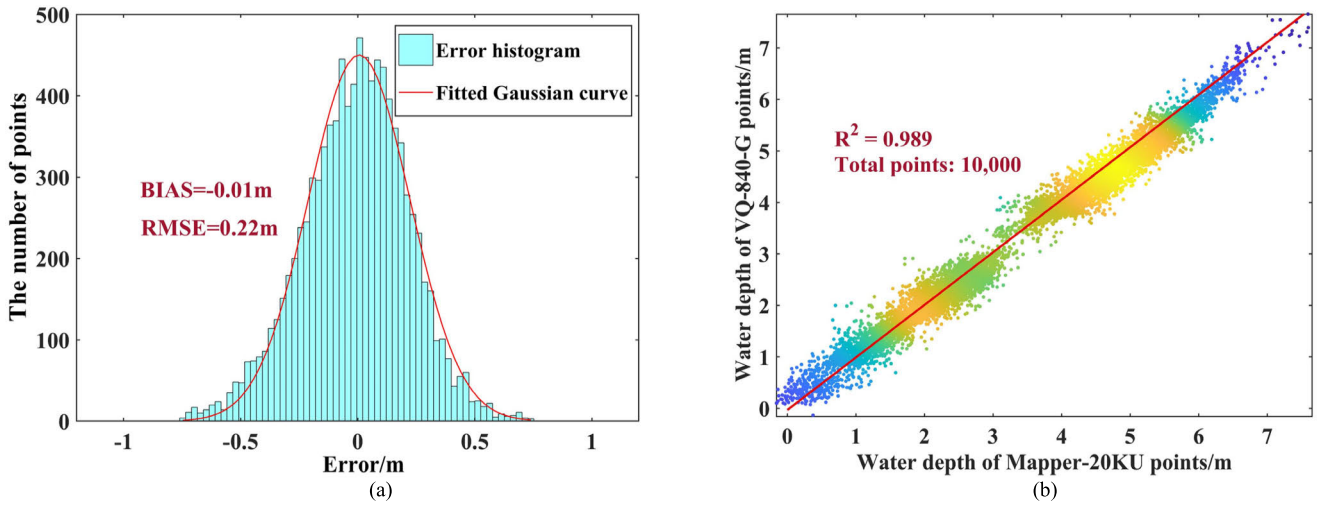


Fig. 12. Quantitative analysis of bathymetric accuracy in Region A. (a) Height difference histogram. (b) Regression analysis against reference data.

water, effectively recovering topographic details of deeper regions. On the basis, two profiles parallel to the flight route were extracted, corresponding to the red and blue dashed lines in Fig. 9(a) and (b). The topographic profiles along the red dashed line are shown in Fig. 9(c) and (d). This profile features significant fluctuations, which are primarily caused by underwater reefs. Furthermore, the topography derived from the SPD algorithm is discontinuous, with interruptions occurring around reefs and greater water depth [see Fig. 9(c)]. These interruptions can be supplemented by the proposed method, as shown in Fig. 9(d). The bathymetric profiles corresponding to the blue dashed line are shown in Fig. 9(e) and (f), where the underwater topography is relatively gentle. The proposed method achieved a maximum depth of 8.67 m in this profile, representing a 52% improvement over the 5.68 m achieved by the SPD method. These results indicate that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

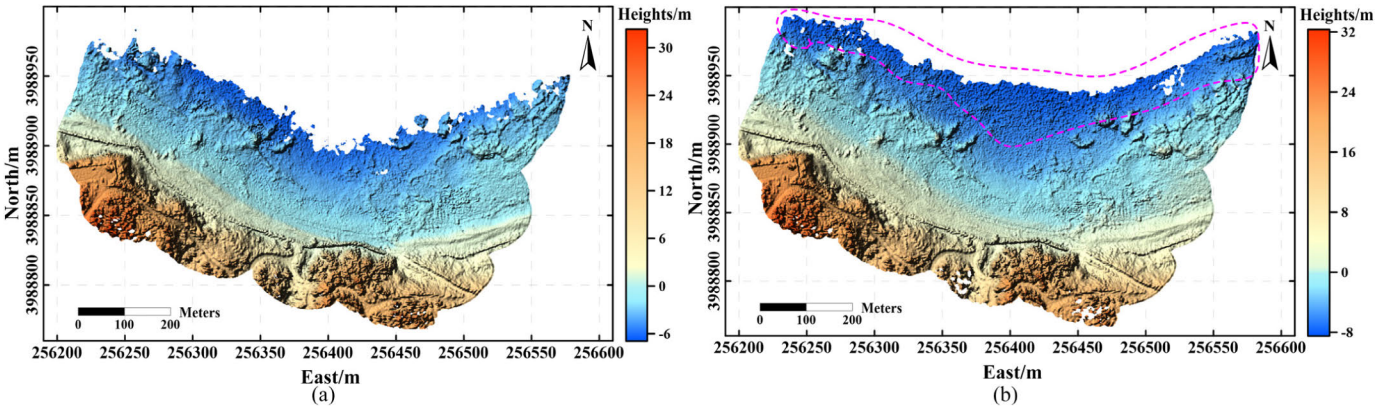


Fig. 13. Water–land integrated topographic data. (a) Topographic data of Region A before processing. (b) Topographic data of Region A after processing.

proposed method can significantly enhance the ALB maximum bathymetric capability in Region A.

In addition, if faint bottom echoes are successfully detected, the number of bottom points will necessarily increase. To evaluate this improvement, the number of bathymetric points at different depths was statistically obtained, as shown in Fig. 10. According to the statistics, within the depth range of 0–3 m, 15 479 and 13 572 bathymetric points were obtained by the proposed method and the SPD method, respectively. It shows only a slight increase. But when the water depth exceeds 3 m, 19 953 bathymetric points can be acquired by the proposed method, while only 2256 bathymetric points can be obtained by the SPD method. After processing with the proposed method, the point density has been raised from 4.6 to 5.1 pts/m^2. These results demonstrate that the proposed method can significantly increase the number of points in deep-water areas, and effectively detect faint bottom echoes.

B. Analysis of Bathymetric Accuracy

In this section, qualitative analysis of bathymetric accuracy is conducted by comparing the topographic profiles between the experimental results and the reference data. The topographic profiles are extracted from the overlapping portions of the two datasets, whose direction is generally perpendicular to the coastline [see Fig. 11(a)]. For easier observation, profile points are interpolated using the inverse distance weighting (IDW) method. It is obvious that the point cloud profiles from two datasets align well, and the topographic trends are generally consistent. Furthermore, profile A of the reference data exhibits discontinuity in its topography, which can be effectively complemented in the experimental result.

For the purpose of quantitative analysis, 10 000 point pairs were randomly selected from the overlapping area between the experimental results and reference data, and a statistical analysis of height differences was performed, as illustrated in Fig. 12(a). The histogram of height differences is observed to approximately follow a Gaussian distribution, which is consistent with the characteristics of a random error distribution. It is noted that the majority of height differences are less than 0.4 m, and the BIAS value is −0.01 m, which indicates that the topographic trends of the two datasets are closely aligned. To further quantify this alignment, the RMSE was calculated as 0.22 m, which satisfies the IHO S-44 Special Order Standard [39]. In addition, a regression analysis of all point pairs is presented in Fig. 12(b). The R^2 value is 0.989, which indicates

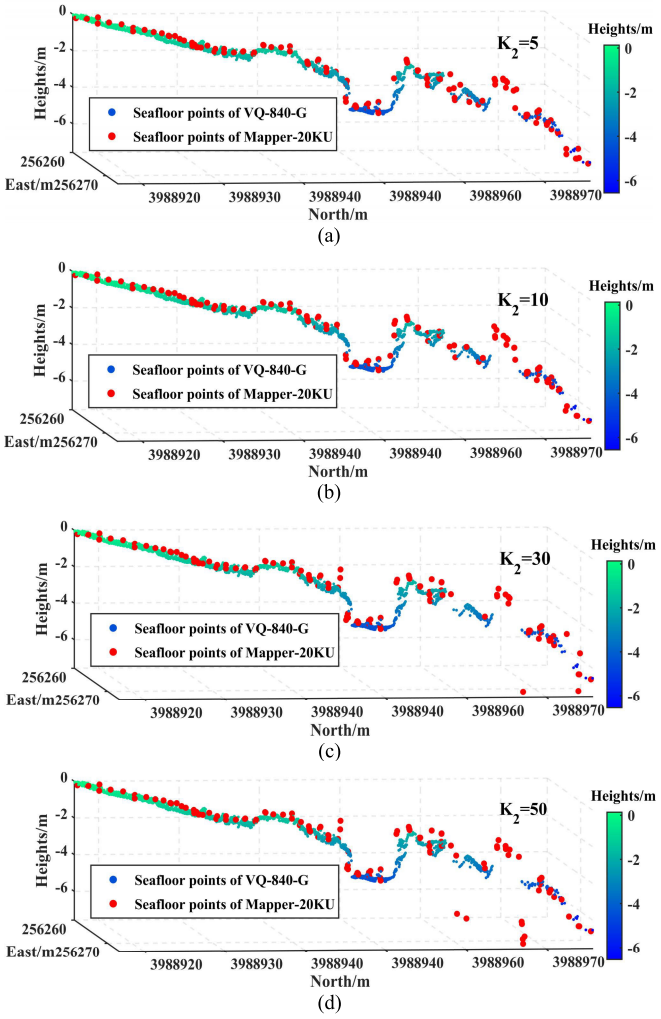


Fig. 14. Seafloor topographic profiles obtained by different K_2 values. (a) Topographic profile points when K_2 equals 5. (b) Topographic profile points when K_2 equals 10. (c) Topographic profile points when K_2 equals 30. (d) Topographic profile points when K_2 equals 50.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

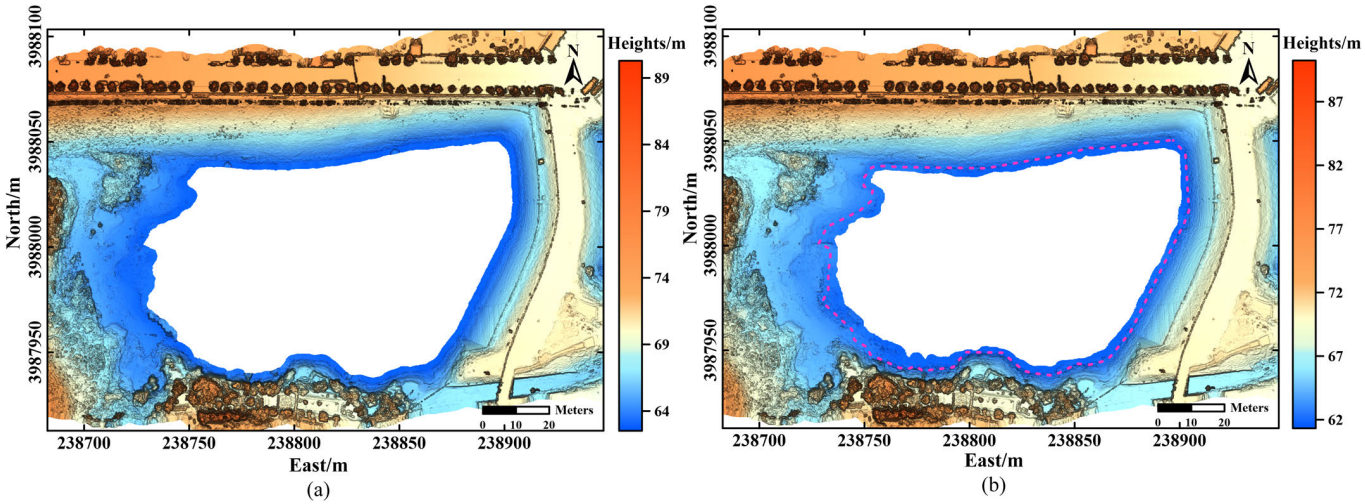


Fig. 15. Water–land integrated topographic data in the reservoir region. (a) Topographic data before processing. (b) Topographic data after processing.

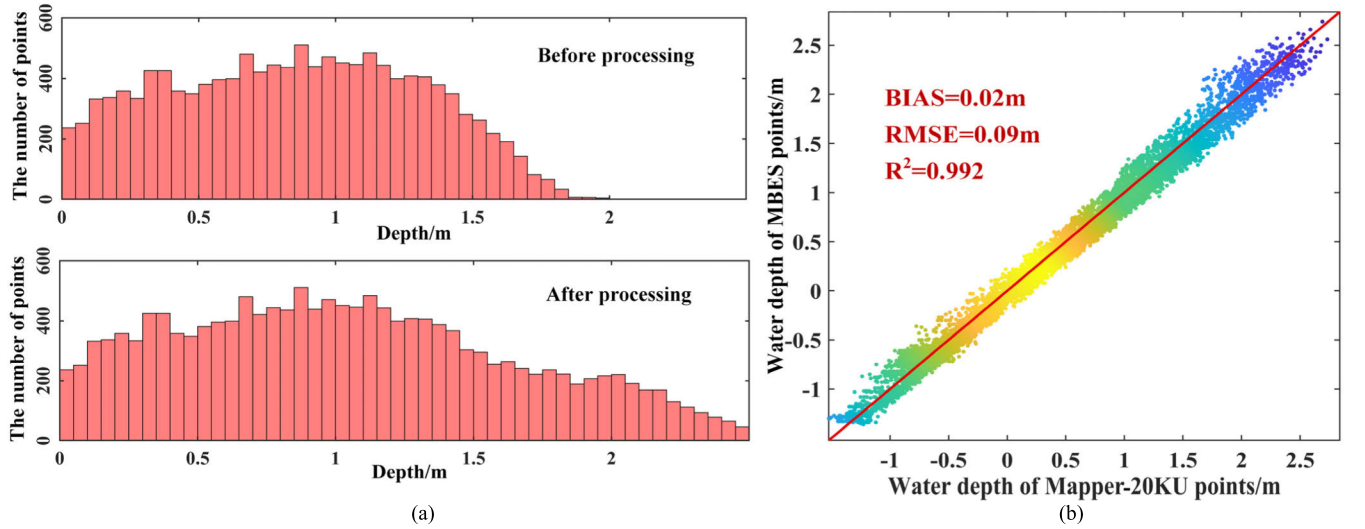


Fig. 16. Bathymetric point statistics and accuracy analysis. (a) Distribution of the number of underwater points. (b) Accuracy analysis against MBES reference data.

a strong correlation between the seafloor points from two datasets, thereby further validating the effectiveness of the proposed method. On this basis, the proposed method was applied to the entire dataset of Region A. The results before and after processing are shown in Fig. 13, where areas of increased water depth are outlined with a pink dashed line.

V. Discussion

A. Analysis of the Influence of K Value on Detection Results

The proposed method utilizes the KNN algorithm to filter spatially neighboring echoes; thus, the K value will inevitably affect the performance of the method. To assess the influence of different K values, profile A in Fig. 11(a) was selected for its typical underwater topographic relief. In this seafloor profile, point clouds were generated for different K values of 5, 10, 30, and 50. These resulting profiles are shown in Fig. 14. By comparing with the reference data, it can be seen that the shallow-water topography is generally consistent, indicating the choice of K_2 value has little impact on the bottom echo detection. In deep-water areas, especially when K_2 is set to 30 or 50, the seafloor points contain many outlier noise points, and the details of the seafloor topography become less distinct [see Fig. 14(c) and (d)], which is caused by the broadening of the pseudo-waveform due to more stacked waveforms. When K_2 is set to 5 or 10 [see Fig. 14(a) or (b)], the topography is more continuous, and the details are clearer. Therefore, to ensure the bathymetric performance, the K_2 value is recommended to be set between 5 and 10, and it is taken as 10 in Section III.

B. Applicability Analysis of the Proposed Method

To assess the applicability of the proposed method in a different water environment, it was applied to the entire inland reservoir dataset shown in Fig. 2(b). The topographic data before and after processing by the proposed method are shown in Fig. 15. Also, areas of increased water depth are outlined with a pink dashed line in Fig. 15(b). Subsequently, the maximum bathymetric depth and accuracy were then evaluated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

in the comparison area marked in Fig. 2(b). Experimental results indicate that the maximum depth is increased from 1.70 to 2.51 m using the proposed method, corresponding to an improvement of approximately 48%. Moreover, statistical analysis [see Fig. 16(a)] shows a significant increase in the number of bathymetric points with depth greater than 1.5 m, and the point density within the comparison area increased from 138 to 156 pts/m^2. In terms of accuracy, 10 000 corresponding point pairs were also randomly selected from the overlapping area of the ALB and MBES point clouds to compute the accuracy indexes. Finally, the results yield a BIAS of 0.02 m, an RMSE of 0.09 m, and an R^2 value of 0.992 [see Fig. 16(b)]. Notably, the RMSE obtained in the reservoir Region B is significantly lower than that observed in coastal Region A (0.22 m). This difference is primarily due to the relative stability of the reservoir environment, whereas the coastal region is influenced by more dynamic factors, including tides, waves, and complex topography. In conclusion, the above results validate that the proposed method effectively improves the maximum bathymetric depth while maintaining the satisfactory accuracy. Therefore, the proposed method is applicable not only in coastal zones but also in inland water bodies.

C. Limitations of the Faint Bottom Echo Detection Method

This study focuses on the determination of ALB faint bottom echoes. Based on the experimental results presented in Section IV, the limitations with corresponding future work can be summarized as follows.

1) As shown in Fig. 11, unavoidable errors remain in the bathymetric results, which are particularly evident in coastal areas where deeper water can be measured. These errors are mainly caused by the complex factors of laser transmission in water, including laser scattering, interaction between laser spots, and irregular topography. Therefore, developing an efficient and accurate error correction method based on the ALB echoes is essential for improving the refinement of underwater topographic data.

2) In addition, in highly turbid water areas with maximum depths of less than 1 m, the ALB bottom echoes are faint and easily get overlapped with surface echoes. The pseudo-bottom echo generated by waveform stacking will further broaden, exacerbating the echo overlap. To address this challenge, it is necessary to develop enhancement algorithms that can highlight bottom echoes and separate overlapped signals.

VI. Conclusion

To achieve accurate faint bottom echo detection of ALB, a constrained waveform stacking model based on dual water-surface elements is proposed in this article. The signal-to-noise ratio of the waveform can be improved based on this model, and then, the bottom echo moment can be determined step-by-step. The method was validated on two datasets from a coastal zone and an inland reservoir, and the results are summarized as follows.

1) Compared with the SPD algorithm, the proposed method increases the maximum depth from 5.68 to 8.67 m in the coastal region and from 1.70 to 2.51 m in the reservoir. Also, the point density in deeper areas is enhanced.

2) In terms of accuracy, the RMSEs of the two regions are 0.22 and 0.09 m, and their correlation coefficients with existing topographic data both exceeded 0.98. It indicates that the faint bottom echoes detected are accurate.

In summary, the proposed method exhibits enhanced adaptability and detection performance for ALB faint bottom echoes, making it applicable not only to coastal zones but also to inland water bodies such as reservoirs and lakes. This study can provide technical support for fields that rely on underwater topography data, including marine engineering and water environment monitoring.

References

[1] P. J. Kinzel, C. J. Legleiter, and J. M. Nelson, "Mapping river bathymetry with a small footprint green LiDAR: Applications and challenges1," *JAWRA J. Amer. Water Resour. Assoc.*, vol. 49, no. 1, pp. 183–204, Feb. 2013.

[2] X. Ji, Z. Dong, Z. Wei, M. Wang, W. Xu, and L. Zhang, "An adaptive local joint-weighted method for continuous stripping intensity correction of airborne bathymetric LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4204112.

[3] L. Wu, Y. Chen, Y. Le, Y. Qian, D. Zhang, and L. Wang, "A high-precision fusion bathymetry of multi-channel waveform curvature for bathymetric LiDAR systems," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103770.

[4] A. Collin, P. Archambault, and B. Long, "Mapping the shallow water seabed habitat with the SHOALS," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 10, pp. 2947–2955, Oct. 2008.

[5] F. Eren, S. Pe'eri, Y. Rzhanov, and L. G. Ward, "Bottom characterization by using airborne LiDAR bathymetry (ALB) waveform features obtained from bottom return residual analysis," *Remote Sens. Environ.*, vol. 206, pp. 260–274, Mar. 2017.

[6] G. Mandlburger, M. Pfennigbauer, R. Schwarz, S. Flöry, and L. Nussbaumer, "Concept and performance evaluation of a novel UAV-borne topo-bathymetric LiDAR sensor," *Remote Sens.*, vol. 12, no. 6, p. 986, Mar. 2020.

[7] E. A. L. Salas, "Waveform LiDAR concepts and applications for potential vegetation phenology monitoring and modeling: A comprehensive review," *Geo-spatial Inf. Sci.*, vol. 24, no. 2, pp. 179–200, Apr. 2021.

[8] S. Xing et al., "A depth-adaptive waveform decomposition method for airborne LiDAR bathymetry," *Sensors*, vol. 19, no. 23, p. 5065, Nov. 2019.

[9] W. Philpot, "Basic concepts in data processing," in *Airborne Laser Hydrography II*. New York, NY, USA: Cornell: ECommons, 2019, pp. 53–146. [Online]. Available: https://www.academia.edu/111777699/Airborne_Laser_Hydrography_II

[10] C. Qi, Y. Ma, D. Su, F. Yang, J. Liu, and X. H. Wang, "A method to decompose airborne LiDAR bathymetric waveform in very shallow waters combining deconvolution with curve fitting," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[11] G. C. Guenther and R. Thomas, "System design and performance factors for airborne laser hydrography," in *Proc. OCEANS*, 1983, pp. 425–430.

[12] G. Zhou, C. Li, D. Zhang, D. Liu, X. Zhou, and J. Zhan, "Overview of underwater transmission characteristics of oceanic LiDAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8144–8159, 2021.

[13] H. Abdallah, N. Baghdadi, J.-S. Bailly, Y. Pastol, and F. Fabre, "WaLiD: A new LiDAR simulator for waters," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 4, pp. 744–748, Jul. 2012.

[14] K. Guo et al., "Development of a single-wavelength airborne bathymetric LiDAR: System design and data processing," *ISPRS J. Photogramm. Remote Sens.*, vol. 185, pp. 62–84, Mar. 2022.

[15] R. Schwarz, G. Mandlburger, M. Pfennigbauer, and N. Pfeifer, "Design and evaluation of a full-wave surface and bottom-detection algorithm for LiDAR bathymetry of very shallow waters," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 1–10, Apr. 2019.

[16] K. Richter, H.-G. Maas, P. Westfeld, and R. Weiß, "An approach to determining turbidity and correcting for signal attenuation in airborne LiDAR bathymetry," *PFG–J. Photogramm., Remote Sens. Geoinf. Sci.*, vol. 85, no. 1, pp. 31–40, Feb. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.

[17] C.-K. Wang, Y.-H. Tseng, and C.-K. Wang, "A wavelet-based echo detector for waveform LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 757–769, Feb. 2016.
[18] X. Liu et al., "Scattering suppression in underwater LiDAR signal processing based on wavelet-ICA method," *IEEE Photon. Technol. Lett.*, vol. 36, no. 5, pp. 345–348, Mar. 2024.
[19] S. Wu, Z. Liu, and B. Liu, "Enhancement of LiDAR backscatters signal-to-noise ratio using empirical mode decomposition method," *Opt. Commun.*, vol. 267, no. 1, pp. 137–144, Nov. 2006.
[20] W. Li et al., "Adaptive denoising for airborne LiDAR bathymetric full waveforms using EMD-based multiresolution analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[21] Z. Zhou et al., "Improvement of the signal to noise ratio of LiDAR echo signal based on wavelet de-noising technique," *Opt. Lasers Eng.*, vol. 51, no. 8, pp. 961–966, Aug. 2013.
[22] X. Zhao, G. Liang, Y. Liang, J. Zhao, and F. Zhou, "Background noise reduction for airborne bathymetric full waveforms by creating trend models using optech CZMIL in the Yellow Sea of China," *Appl. Opt.*, vol. 59, no. 35, pp. 11019–11026, Dec. 2020.
[23] C. Liu, X. Li, J. Huang, and L. Xu, "B-spline based progressive decomposition of LiDAR waveform with low SNR," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.
[24] T.-A. Teo and W.-Y. Yeh, "The benefit of the geospatial-related waveforms analysis to extract weak laser pulses," *Remote Sens.*, vol. 10, no. 7, p. 1141, Jul. 2018.
[25] X. Li and P. Luo, "Boosting ranging performance of LiDAR using multi-pulse coherent average," *IEEE Sensors J.*, vol. 19, no. 15, pp. 6270–6278, Aug. 2019.
[26] Y. Huang, Y. He, X. Zhu, J. Yu, and Y. Chen, "Faint echo extraction from ALB waveforms using a point cloud semantic segmentation model," *Remote Sens.*, vol. 15, no. 9, p. 2326, Apr. 2023.
[27] W. Yao and U. Stilla, "Mutual enhancement of weak laser pulses for point cloud enrichment based on full-waveform analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3571–3579, Sep. 2010.
[28] D. Mader, K. Richter, P. Westfeld, and H.-G. Maas, "Potential of a non-linear full-waveform stacking technique in airborne LiDAR bathymetry: Demonstration of full-waveform stacking techniques on data from the elbe river," *PFG – J. Photogramm., Remote Sens. Geoinformation Sci.*, vol. 89, no. 2, pp. 139–158, Jun. 2021.
[29] D. Mader, K. Richter, P. Westfeld, J.-G. Nistad, and H.-G. Maas, "Analysis of the potential of full-waveform stacking techniques applied to coastal airborne LiDAR bathymetry data of the German Wadden sea national park," *Int. Hydrographic Rev.*, vol. 29, no. 2, pp. 46–64, Nov. 2023.
[30] D. Mader, K. Richter, P. Westfeld, and H.-G. Maas, "Volumetric nonlinear ortho full-waveform stacking in airborne LiDAR bathymetry for reliable water bottom point detection in shallow waters," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 145–162, Oct. 2023.
[31] M. Pfennigbauer, C. Wolf, J. Weinkopf, and A. Ullrich, "Online waveform processing for demanding target situations," *Proc. SPIE*, vol. 9080, Jun. 2014, Art. no. 90800J.
[32] R. Schwarz, N. Pfeifer, M. Pfennigbauer, and A. Ullrich, "Exponential decomposition with implicit deconvolution of LiDAR backscatter from the water column," *PFG–J. Photogramm., Remote Sens. Geoinf. Sci.*, vol. 85, no. 3, pp. 159–167, Aug. 2017.
[33] B. Tao et al., "Precise detection of water surface through the analysis of a single green waveform from bathymetry LiDAR," *Opt. Exp.*, vol. 30, no. 22, pp. 40820–40841, Oct. 2022.
[34] F. Yang et al., "Modeling and analyzing water column forward scattering effect on airborne LiDAR bathymetry," *IEEE J. Ocean. Eng.*, vol. 48, no. 4, pp. 1373–1388, Oct. 2023.
[35] P. Westfeld, H.-G. Maas, K. Richter, and R. Weiß, "Analysis and correction of ocean wave pattern induced systematic coordinate errors in airborne LiDAR bathymetry," *ISPRS J. Photogramm. Remote Sens.*, vol. 128, pp. 314–325, Jun. 2017.
[36] D. Su, F. Yang, Y. Ma, X. H. Wang, A. Yang, and C. Qi, "Propagated uncertainty models arising from device, environment, and target for a small laser spot airborne LiDAR bathymetry and its verification in the South China Sea," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3213–3231, May 2020.
[37] J. Sanchez, F. Denis, D. Coeurjolly, F. Dupont, L. Trassoudaine, and P. Checchin, "Robust normal vector estimation in 3D point clouds through iterative principal component analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 18–35, May 2020.
[38] W. Xu et al., "Refraction error correction of airborne LiDAR bathymetry data considering sea surface waves," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102402.
[39] (2020). *IHO Standards for Hydrographic Surveys*. [Online]. Available: https://www.hydro.navy.mi.th/standard/S-44_Edition_6_Final.pdf

Xiaolin Yu received the B.S. degree in geodesy and geomatics from Shandong University of Science and Technology, Qingdao, China, in 2020, where he is currently pursuing the Ph.D. degree with the College of Geodesy and Geomatics.

His research interests include airborne LiDAR bathymetric data processing.

Dianpeng Su received the Ph.D. degree in ocean geomatics from Shandong University of Science and Technology, Qingdao, China, in 2018.

He is currently with the College of Geodesy and Geomatics, Shandong University of Science and Technology. His research interests include bathymetry surveys and airborne LiDAR bathymetry processing.

Fanlin Yang received the Ph.D. degree in geodesy and geomatics from Wuhan University, Wuhan, Hubei, China, in 2003.

He is currently a Professor with the College of Geomatics, Shandong University of Science and Technology, Qingdao, China. His research areas focus on bathymetry survey and marine navigation.

Wenze Yang received the B.S. degree in geodesy and geomatics from North China University of Science and Technology, Tangshan, China, in 2024. He is currently pursuing the master's degree with the College of Geodesy and Geomatics, Shandong University of Science and Technology, Qingdao, China.

His research interests include the processing of waveform data from airborne LiDAR bathymetry.

Zhenhui Wang received the master's degree in geodesy and surveying engineering from Wuhan University, Wuhan, Hubei, China, in 2006.

He is currently with the Seventh Topographic Surveying Bridge, Ministry of Natural Resources of China, Qingdao, China. His research interests include geodesy and GNSS applications research.

Chunxiao Wang received the Ph.D. degree in cartography and geographic information engineering from Shandong University of Science and Technology, Qingdao, China, in 2012.

He is currently with Hainan Basic Geographic Information Centre, Ministry of Natural Resources of China, Qingdao, China. His research interests include marine surveying technology and engineering surveying applications.

Anxiu Yang received the Ph.D. degree in marine surveying and mapping from Shandong University of Science and Technology, Qingdao, China, in 2021.

She is currently a Lecturer with the College of Geodesy and Geomatics, Shandong University of Science and Technology. Her research interests include LiDAR point cloud processing and multisource data fusion.

Chao Qi received the Ph.D. degree in ocean geomatics from Shandong University of Science and Technology, Qingdao, China, in 2022.

He is currently with the College of Geodesy and Geomatics, Shandong University of Science and Technology. His research interests include bathymetry survey and airborne LiDAR bathymetry processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:46:00 UTC from IEEE Xplore. Restrictions apply.